

# AAGAM

## Complete Project Lifecycle, System Features and Prototype Report

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*Adaptive multi-model weather forecast blending and decision-support platform*

**Smart India Hackathon 2026 | Problem Statement 26081 | MoES / NCMRWF**

### Prototype status

Operational demonstration build. Current public prototype is designed for research, forecasting support and demonstration use across 40 configured locations in India. It is not an official warning or disaster bulletin system.

Repository: <https://github.com/bitsubhayu/AAGAM>

Live prototype: <https://aagam-mlb8.vercel.app>

Report date: 26 September 2026

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## 1. Executive Overview

AAGAM is a weather forecasting and decision-support platform that combines four independent forecast sources, measures their recent performance, creates an adaptive blend, and presents the result through an operational dashboard. **The central idea is simple: rather than trusting one model equally everywhere, AAGAM measures which source has been performing better for a particular variable, lead time, region and season, then adjusts the contribution of each source accordingly.**

### 1.1 Core value delivered

- A single blended forecast for rainfall, maximum temperature and wind for Day 0 through Day 7.
- Transparent regional and seasonal model weights rather than a hidden ensemble.
- Forecaster-controlled weight overrides with mandatory justification and an audit trail.
- Rule-based extreme weather guidance using rainfall, heat, wind, multi-day rainfall and uncertainty signals.
- Continuous verification against observation data, with model-by-model comparisons and an explicit equal-weight baseline.
- Role-based access for public users, forecasters and a Forecaster Coordinator.
- Email-based alert subscriptions so users can receive important changes without continuously watching the dashboard.
- Research access to historical records, raw model values and downloadable CSV/JSON datasets.
- Automated data ingestion, verification, retraining, model evaluation, backup and version switching.

### 1.2 Current prototype position

The current build is a complete end-to-end prototype rather than a single isolated forecasting component. It includes the data layer, scoring and blending engines, operational scheduler, secure API, browser interface, alert lifecycle, subscriptions, verification views and research/export workflows. The current production branch is synchronized at commit 9d52b408f11ed067e3caf6095c26575c08eadccc.

## 2. Problem Context and AAGAM Objectives

Different numerical weather prediction systems perform differently under different weather regimes, lead times and geographical conditions. AAGAM was designed to reduce the need for manual side-by-side comparison by turning model performance history into a repeatable blending process.

### 2.1 Objectives

- Improve the practical usefulness of multi-model forecasts by adapting source contributions to recent verified performance.
- Make model trust visible through weights, verification metrics and model-by-model comparison charts.
- Generate early decision-support flags for selected extreme weather conditions.
- Keep routine operation unattended through scheduled jobs and automated checks.
- Preserve raw and historical data so forecasters and researchers can reproduce or investigate decisions.

### 2.2 Important positioning

#### Decision support, not an official bulletin

AAGAM is a research and demonstration platform. Its forecasts and alert flags must not be treated as the official Government of India, IMD or disaster-management warning product. Official warnings and emergency instructions remain authoritative.

### 3. What the Prototype Can Do Today

| Capability          | Prototype implementation                                                               |
|---------------------|----------------------------------------------------------------------------------------|
| Forecasting         | 4-source blend; rainfall, Tmax and wind; Day 0-7; 40 locations                         |
| Weight Maps         | 5-region x 7-lead matrix plus per-location dominant source map                         |
| Forecaster Override | Audited, temporary adjusted blend with reason and expiry                               |
| Verification        | MAE, RMSE, bias and rainfall POD/FAR/CSI; regional and seasonal views                  |
| Extreme Weather     | Heavy rain, heatwave, high wind, 3-day heavy rain and high uncertainty                 |
| Alerts              | Severity, model agreement, spread, lifecycle tracking and acknowledgement              |
| Subscriptions       | Email OTP login; saved locations/hazards/severity; daily summary and lifecycle mail    |
| Assistant           | Explain / Raw / Both modes; forecast, weights, skill, alerts, history and export tools |
| Research Access     | CSV/JSON export, raw model records, observed data and history queries                  |
| Operations          | Scheduled ingestion, verification, weekly retraining, backup and health monitoring     |

### 4. End-to-End System Lifecycle

Development was structured as a sequence of connected layers. Each stage depended on the outputs of the previous stage and introduced its own checks before the next layer was opened.

| Stage   | Focus            | What was delivered                                                                                                         |
|---------|------------------|----------------------------------------------------------------------------------------------------------------------------|
| Phase 0 | Foundation       | Repository structure, environment, database, configuration, testing and deployment foundations.                            |
| Phase 1 | Data foundation  | 40 locations, four forecast sources, historical backfill, observation data, IST-day aggregation and Parquet training data. |
| Phase 2 | Skill scoring    | MAE/RMSE/bias, equal-mean and best-single baselines, inverse-error skill weights and hierarchical fallback.                |
| Phase 3 | Blending engines | Positive Ridge and LightGBM, time-based validation, blend selection and evaluation reports.                                |
| Phase 4 | Extreme guidance | Rainfall, heat, wind and uncertainty rules plus rainfall categorical verification.                                         |
| Phase 5 | Live operations  | Scheduled ingestion, verification, retraining, retention, model registry, backups and pipeline health.                     |
| Phase 6 | API layer        | FastAPI endpoints, JWT validation, role checks, RLS-aware database access, rate limits and response contracts.             |

|                 |                            |                                                                                                                             |
|-----------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Phase 7         | Dashboard                  | Overview, Forecast Explorer, Weight Maps, Skill, Extreme Weather, Data & Export, Settings and operational states.           |
| Phase 8         | AAGAM Assistant            | Grounded tool access, structured tables, Explain/Raw/Both modes, streaming responses and numeric checks.                    |
| Phase 9         | Hardening                  | Reliability tests, backup/rollback drills, documentation, performance checks and demo readiness.                            |
| Phase 10        | Public access and events   | Public read model, alert-event grouping, lifecycle states and public event detail views.                                    |
| Phase 11        | Notifications              | Email OTP, subscriptions, daily summary and alert lifecycle notifications.                                                  |
| Phase 12        | Trust and context          | Alert track record, plain-language guidance, shareable event pages and event outcomes.                                      |
| Phase 13        | Climatology                | Location-specific percentiles, rarity context and 3-day cumulative rainfall guidance.                                       |
| Final hardening | Production synchronization | Resolved live/frontend contract issues, verified Weight Maps, synchronized main branch and confirmed production deployment. |

## 5. Forecast Inputs, Truth Data and Coverage

### 5.1 Forecast sources

| Source     | Provider | Role in AAGAM                                         |
|------------|----------|-------------------------------------------------------|
| GFS        | NOAA     | Global physical forecast source                       |
| ECMWF IFS  | ECMWF    | Global physical forecast source at 0.25 degree        |
| DWD ICON   | DWD      | Independent global physical forecast source           |
| ECMWF AIFS | ECMWF    | Independent forecast source used as the fourth member |

### 5.2 Verification truth

- Rainfall: IMD gridded rainfall at 0.25 degree, accessed through the project data workflow.
- Maximum temperature and wind: ERA5-based historical data through the historical weather workflow.
- Recent truth availability is tracked because observation products have a publication lag.
- Every truth row retains its source so verification is traceable.

### 5.3 Time conventions

- Storage uses UTC.
- Rainfall is aggregated to the 08:30 IST to 08:30 IST day boundary.
- Tmax and daily wind maxima use the IST calendar day convention.
- Lead times are tracked per valid date, with Day 0 included in the current operational API and Day 1-7 used for standard forward-looking views.

## 5.4 Coverage

The prototype covers 40 representative Indian locations grouped into five operational regions: North-West (NW), Central, East & North-East (EAST\_NE), South Peninsula (SOUTH), and Himalayan/Hilly (HIMALAYAN). Each location is classified as plains, coastal or hills where relevant to threshold handling.

## 6. Forecasting and Blending Method

AAGAM uses a transparent baseline plus two blending engines. The goal is to make the final blend both useful and inspectable rather than treating the ensemble as an opaque average.

### 6.1 Transparent skill-weight baseline

For a chosen variable, lead, region and season bucket, the source weights are proportional to the inverse of recent MAE. The weights are then normalized to sum to 1. This baseline is also the reference shown in the Weight Maps feature.

### 6.2 Ridge blend

Ridge learns a non-negative mixture of the four source forecasts. The non-negative constraint keeps the output interpretable as a contribution breakdown, which is why Ridge is used directly for the Weight Maps and forecaster override workflow.

### 6.3 LightGBM blend

LightGBM uses the four source forecasts together with contextual features such as mean, spread, extrema, lead time, day-of-year terms, location and region. It is used to capture non-linear relationships that a fixed weighted sum may miss.

### 6.4 Final blend selection

- For each variable and lead time, Ridge and LightGBM are compared on validation error.
- The lower-error engine is selected for the final blend.
- When their validation error is within about 2 percent, the two outputs are averaged.
- The system also reports equal-weight and best-single baselines so performance can be examined honestly.

#### Simple view of the blending chain

Four forecast sources -> recent skill measurement -> adaptive weights / fitted blending -> final blended value -> verification + extreme-weather rules -> dashboard / alerts / research export.

## 7. Automatic Model-Version Switching

Model versions are treated as managed candidates rather than being replaced immediately after training. The active version stays protected while a new candidate is evaluated.

### 7.1 Simple algorithm

1. Train a new candidate version on the latest eligible history.
2. Evaluate the candidate against the currently active version on a time-separated validation set.
3. Apply a promotion margin that accounts for sample size and evaluation consistency.
4. Move a qualifying candidate through staged observation: evaluating -> eligible -> shadow -> canary -> active.
5. Keep the current version if the candidate fails any gate; automatically roll back if a promoted candidate later fails operational safety checks.

## 7.2 Promotion margin

**Required margin = BASE\_MARGIN x (1 + sample-factor + inconsistency-factor).** The purpose is to prevent a candidate from being promoted because of a tiny or unstable apparent improvement. For example, a change such as 0.812 to 0.814 is intentionally treated as insufficient when it does not clear the required margin.

## 7.3 Staged safety

- Shadow evaluation persists the candidate before live exposure.
- Canary evaluation uses a small set of locations, one per operational region.
- Promotion and rollback are recorded in the version decision and audit records.
- Routine user operation does not require a person to manually switch the production version.

# 8. Weight Maps and Forecaster Overrides

## 8.1 Weight Maps

- The main heatmap is organized as 5 regions x 7 lead days.
- Each cell shows the dominant source, the strength of dominance and the underlying model weights.
- The detail inspector shows the four model contributions, sample count and the fallback level used when a fully specific bucket is not available.
- A second map view displays the dominant source for each of the 40 locations at the selected lead time.

## 8.2 Forecaster override workflow

1. A forecaster selects a region, season, variable and lead-time cell.
2. The current four-source weights are loaded into an override form.
3. The sliders are normalized so the final contributions sum to 100 percent.
4. A written meteorological reason is mandatory.
5. An expiry can be selected for the temporary adjustment.
6. The adjusted blend is stored separately and shown with an audit record; the official baseline blend is not overwritten.

### Governance rule

Weight overrides are reversible, time-bounded and traceable to the responsible forecaster. They are intended to capture expert operational judgment without rewriting the historical baseline.

# 9. Role-Based Access and User Responsibilities

| Role                   | Responsibilities                                                                                                                              |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Public                 | View forecasts, maps, weights, skill, alerts, history and exports; use the assistant; manage a personal alert subscription after OTP sign-in. |
| Forecaster             | All public capabilities plus acknowledge alerts, apply audited weight overrides and inspect operational pipeline information.                 |
| Forecaster Coordinator | All forecaster capabilities plus manage verified forecaster promotion to coordinator.                                                         |

## 9.1 Access model

- Core read pages are available publicly without a login.
- Email OTP authentication is used for notification subscriptions and forecaster operational actions.
- Database Row Level Security enforces row-level permissions.
- The service-role database credential remains server-side and is never sent to the browser.
- Anonymous access and authenticated access have separate rate limits.

## 10. AAGAM Assistant

AAGAM Assistant provides a single conversational entry point to the same verified operational data that powers the dashboard. It can return either exact tables, plain-language explanations or both, depending on the selected mode.

### 10.1 Modes

| Mode    | Output                                                                           |
|---------|----------------------------------------------------------------------------------|
| Explain | 2-5 sentence plain-language explanation with units, dates and data-used context. |
| Raw     | Sortable data table with download options; no extended interpretation.           |
| Both    | Exact table plus a short interpretation.                                         |

### 10.2 What the assistant can retrieve

- Latest forecast for one configured location and variable, including each source value, the blend, spread and threshold agreement.
- Weight matrices for a selected region or location, season and lead.
- Skill scores by lead, region, season or model.
- Current or historical alerts with severity, agreement and rule inputs.
- Historical forecast and observed records over a selected period.
- Downloadable datasets in CSV or JSON format.

### 10.3 Data integrity controls

- The assistant is restricted to a fixed tool set; it does not generate arbitrary database queries.
- Numbers shown in the response are checked against the corresponding tool result.
- Raw table data is delivered from the backend artifact store rather than reconstructed from prose.
- The assistant respects user role, request limits and the 40-location coverage boundary.

## 11. Extreme Weather Guidance and Alerting

Alerts are generated automatically during the forecast cycle. They combine forecast values, multi-source agreement and configurable rule thresholds. This allows users to receive meaningful changes without watching the dashboard continuously.

| Signal     | Current rule summary                                                                                                                                                                                    |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Heavy rain | Advisory: any source $\geq 64.5$ mm or blend $\geq 80\%$ of threshold; Watch: blend $\geq 64.5$ mm or at least 2 sources $\geq 64.5$ ; Alert: blend $\geq 64.5$ mm and at least 3 sources $\geq 64.5$ . |

|                  |                                                                                                                                                                                           |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Heatwave         | Uses terrain-specific Tmax thresholds and departure from normal; condition is required for 2 consecutive forecast days. Absolute thresholds also cover $\geq 45$ C and $\geq 47$ C cases. |
| High wind        | Advisory $\geq 50$ km/h; Watch $\geq 62$ km/h; Alert $\geq 75$ km/h; storm threshold $\geq 89$ km/h.                                                                                      |
| 3-day heavy rain | Location-specific p90 / p95 / p99 cumulative rainfall guidance when at least 15 years of truth history are available.                                                                     |
| High uncertainty | Flag when model spread exceeds the historical 90th percentile for the corresponding bucket.                                                                                               |

### 11.1 Alert lifecycle

- Consecutive flagged periods are grouped into alert events.
- Individual alert records carry lifecycle states such as new, upgraded, downgraded, unchanged and cancelled.
- Events that pass beyond their validity become expired; events that drop out before arrival can be marked cancelled.
- Cancelled records are retained with the responsible name and audit information rather than deleted.
- Forecasters can acknowledge relevant alerts; the acknowledgement is recorded.

### 11.2 Email notification workflow

- A visitor enters an email address and completes a 6-digit OTP flow.
- The subscriber selects locations, hazard types and minimum severity.
- A daily summary can be delivered at the configured morning schedule.
- Lifecycle emails can report meaningful alert changes such as new, upgraded or downgraded events.
- Notification logs support deduplication and debugging so unchanged cycles do not repeatedly send the same message.

#### Operational benefit

A user does not need to sit in front of the dashboard throughout the day. The dashboard remains the full decision surface, while subscriptions provide a lightweight notification channel for important changes.

## 12. Forecast Explorer, Verification and Skill Reporting

### 12.1 Forecast Explorer

- Searchable location selector.
- Variable and lead-day controls.
- Four source lines plus the thick blended line.
- Min-max envelope and spread information.
- IMD threshold overlays where applicable.
- Exact-value table with copy and download support.
- Direct hand-off to the assistant for questions about the selected forecast.

### 12.2 Verification and skill

- Continuous verification stores MAE, RMSE and bias for the four sources plus equal-mean, Ridge, LightGBM and the final blend.
- Rainfall verification also uses POD, FAR and CSI at configured rainfall cutoffs.
- Skill can be viewed by lead time, region and season.

- The formal held-out benchmark is separated from the rolling live verification window.
- The current formal benchmark definition uses the fixed period 21 June 2026 through 18 September 2026.
- The live view uses only the verifiable operational window, capped at 90 days, and labels short windows as preliminary or early rather than presenting them as mature statistics.
- All-India aggregation is computed from pooled observation records rather than averaging regional percentages.

### Honest reporting rule

The Skill page is designed to show where the blend improves performance and where it does not. Weak cells are not hidden just to produce a cleaner presentation.

## 13. Research Data, Raw Model Access and Export

AAGAM is also intended to be useful as a research workstation. Forecast consumers can inspect what each source actually produced instead of seeing only the final blend.

- Historical queries can return forecast, observed or blended records.
- Raw records retain the source values for GFS, ECMWF IFS, ICON and AIFS where available.
- Researchers can compare source forecasts with observed values over a chosen time range.
- The Data & Export page provides CSV/JSON export for forecast, history, skill, weights and alerts.
- The assistant can prepare the same datasets in Raw mode, then expose a backend-generated download link.
- Large assistant history responses are intentionally capped; broader research extracts should use the export workflow.

## 14. Main Application Screens and User Workflow

| Screen               | Purpose                                                                                                     |
|----------------------|-------------------------------------------------------------------------------------------------------------|
| Overview             | National situation view, KPIs, alerts, freshness state, map and access to selected points.                  |
| Forecast Explorer    | Compare the four source forecasts, blend, envelope and threshold lines for a location.                      |
| Weight Maps          | Inspect regional/seasonal/lead weighting and the dominant source per station; forecaster override workflow. |
| Skill & Verification | Review model and blend performance, test period, live verification and rainfall categorical scores.         |
| Extreme Weather      | Filter and inspect alerts, why-flagged details, lifecycle, rarity, guidance, track record and sharing.      |
| AAGAM Assistant      | Conversation, data tables, explanation, raw mode, citations/data-used context and downloads.                |
| Data & Export        | Dataset selection, filters, preview and CSV/JSON download for research work.                                |
| Pipeline Health      | Recent jobs, status, rows, API-call estimates, model versions and operational freshness.                    |
| Settings / Access    | Role information, forecaster-access workflow and user configuration where applicable.                       |
| Public Alert Share   | Read-only public event page for sharing a specific alert event                                              |

without requiring login.

## 15. Automation, Reliability, Security and Data Governance

### 15.1 Scheduled operations

| Job                        | Cadence       | Purpose                                                                                            |
|----------------------------|---------------|----------------------------------------------------------------------------------------------------|
| Ingest + blend             | Every 6 hours | Fetch new source forecasts, aggregate to IST days, blend, flag extremes and write current results. |
| Daily verification         | Daily         | Match now-verifiable forecasts to truth and refresh verification records.                          |
| Weekly retraining          | Weekly        | Retrain, evaluate, register candidate version and apply the automatic version gate.                |
| Nightly backup             | Nightly       | Export key operational tables to Parquet in protected storage.                                     |
| Daily notification summary | Morning       | Send daily digest to active subscribers according to saved preferences.                            |
| Climatology maintenance    | Periodic      | Update location-specific truth-based percentiles for local extremeness context.                    |
| Continuous CI              | Push / PR     | Run linting, tests and production-build checks.                                                    |

### 15.2 Resilience features

- Exponential backoff and retry for upstream requests.
- Clean handling of API rate limits and partial source availability.
- Hierarchical fallback for sparse skill buckets instead of inventing noisy weights.
- Missing values remain missing; the system does not silently substitute a value from another source.
- Stale-data banner when the operational data freshness limit is exceeded.
- Cold-start state when the backend service needs time to wake up.
- Separate error states for page sections so one failed API request does not have to blank the whole dashboard.

### 15.3 Security and governance

- Supabase JWT validation on protected API calls.
- Role resolution from verified user profiles.
- Row Level Security on application data tables.
- Server-only use of privileged database credentials.
- CORS restricted to configured application domains.
- Anonymous and authenticated rate limits.
- Append-style operational and assistant audit records with retention rules.

## 16. Development and Hardening History

The prototype was hardened through repeated integration checks, not only through isolated component tests. Several important issues were identified late in the lifecycle and then converted into permanent safeguards.

| Area                       | Issue found                                                                                       | Permanent response                                                                                 |
|----------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Forecast cycle selection   | The API could return more than one issue cycle and the UI could confuse Day 0 with Day 1.         | Latest issue cycle isolation and explicit Day 0 preservation.                                      |
| Skill verification window  | The live verification window and the actual verifiable data did not always match.                 | Dynamic effective window with maturity labels and a separate fixed held-out benchmark.             |
| Assistant data grounding   | Some answers could fall back to local test data or infer values that were not current.            | Controlled production data paths, canonical knowledge, unified skill service and numeric checking. |
| Map reliability            | Map basemap initialization could fail independently of data markers.                              | Load/failure watchdog, cleanup and OSM fallback.                                                   |
| Weight Maps blank page     | Frontend override fields did not match the live backend response shape.                           | Contract alignment, safe rendering, dedicated error boundaries and regression tests.               |
| Pipeline test flakiness    | Retention and verification tests could interfere through shared database state.                   | Scoped cleanup, self-healing fixtures and pooler-resilient connection handling.                    |
| Production synchronization | Verified fixes can exist on a feature branch while production still serves the older main branch. | Explicit branch comparison, fast-forward synchronization and live bundle verification.             |

## 17. Current Technology and Operational Resources

| Area                  | Resources / technologies                                                                                                                       |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Frontend              | React, TypeScript, Vite, Tailwind CSS, shadcn/ui, Apache ECharts, Leaflet/react-leaflet, TanStack Query, Zustand, Sonner, react-markdown, Zod. |
| Backend               | Python, FastAPI, Uvicorn, asyncpg, SlowAPI, Supabase JWT validation and structured JSON APIs under /api/v1.                                    |
| Scientific processing | pandas, NumPy, xarray, PyArrow, scikit-learn Ridge, LightGBM and imdlib.                                                                       |
| Database / storage    | Supabase PostgreSQL with PostGIS, Row Level Security and Storage buckets for data, model artifacts and backups.                                |
| Automation            | GitHub Actions for scheduled ingestion, verification, retraining, backup, notifications and CI.                                                |
| Email                 | Brevo SMTP/API for OTP and notification delivery.                                                                                              |
| Deployment            | Vercel for the web application and Render for the FastAPI backend.                                                                             |
| Source repository     | GitHub repository with documented PRD, technology specification, design system, configuration and test suites.                                 |

### 17.1 Operational architecture in one line

Forecast sources + truth -> ingestion -> validation/aggregation -> skill scoring -> blending -> version registry -> database/storage -> FastAPI -> dashboard / alerts / assistant / export

## 18. Current Scope, Boundaries and Limitations

- Coverage is limited to 40 configured points rather than a continuous national grid.

- The active forecasting scope is rainfall, maximum temperature and maximum wind.
- The system is not a radar, satellite nowcasting or independent numerical weather model.
- Direct cyclone-track forecasting, atmospheric soundings and a full hydrological flood model are outside the current scope.
- AAGAM provides decision-support flags and does not issue official IMD warnings.
- Upstream data delays can create stale or partial cycles, and these conditions are surfaced rather than hidden.
- The demonstration environment uses limited service quotas, so a future operational deployment would require capacity planning and production-grade service plans.

### 18.1 What AAGAM predicts now

The current prototype directly produces numeric forecasts for:

- 24-hour accumulated rainfall (mm/24h).
- Maximum temperature (degrees C).
- Maximum wind speed / gust representation (km/h).

From those variables it derives decision-support guidance for heavy rain, heatwave conditions, high wind, prolonged 3-day rainfall and high uncertainty.

## 19. Future Extension Path

- Expand beyond 40 representative points toward a denser national grid when compute and storage capacity allow.
- Add more variables such as Tmin, subject to verified source data and a clear evaluation plan.
- Integrate additional ensemble information for richer spread and confidence analysis.
- Add a formal public machine-readable alert feed after appropriate governance review.
- Provide broader advisory export formats for operational briefings.
- Move from demonstration quotas to production-grade service limits for institutional deployment.

## 20. Final Prototype Status and Reference Links

| Item                             | Current status / reference                                                            |
|----------------------------------|---------------------------------------------------------------------------------------|
| Current main commit              | 9d52b408f11ed067e3caf6095c26575c08eadccc                                              |
| Weight Maps status               | Production verified; blank-page crash resolved.                                       |
| Frontend checks                  | 0 lint errors; production build completed successfully.                               |
| Weight override regression suite | 17 passed, 0 failed.                                                                  |
| Production frontend              | <a href="https://aagam-mlb8.vercel.app">https://aagam-mlb8.vercel.app</a>             |
| GitHub repository                | <a href="https://github.com/bitsubhayu/AAGAM">https://github.com/bitsubhayu/AAGAM</a> |

### Reference documents used for this report

- AAGAM\_PRD.md - product requirements and functional scope
- AAGAM\_TECH\_STACK.md - technologies, data sources, infrastructure and schedules
- PRODUCT.md - product definition and user foundations
- DESIGN.md - interface and design system specification
- docs/ARCHITECTURE.md - technical architecture
- docs/LIMITATIONS.md - operational boundaries and disclaimers

- config/locations.yaml - authoritative 40-location list
- config/thresholds.yaml - alert and threshold rules
- config/hazard\_guidance.yaml - team-authored hazard guidance
- Production synchronization and Weight Maps verification reports from September 2026

**GitHub:** <https://github.com/bitsubhayu/AAGAM>

**Live prototype:** <https://aagam-mlb8.vercel.app>

## Appendix A. Complete 40-Location Coverage List

### EAST\_NE (10)

|                       |                        |                      |
|-----------------------|------------------------|----------------------|
| Kolkata - West Bengal | Basirhat - West Bengal | Bhubaneswar - Odisha |
| Guwahati - Assam      | Siliguri - West Bengal | Patna - Bihar        |
| Ranchi - Jharkhand    | Shillong - Meghalaya   | Dibrugarh - Assam    |
| Agartala - Tripura    |                        |                      |

### SOUTH (8)

|                                |                       |                             |
|--------------------------------|-----------------------|-----------------------------|
| Chennai - Tamil Nadu           | Kochi - Kerala        | Thiruvananthapuram - Kerala |
| Visakhapatnam - Andhra Pradesh | Hyderabad - Telangana | Bengaluru - Karnataka       |
| Mangaluru - Karnataka          | Madurai - Tamil Nadu  |                             |

### CENTRAL (7)

|                       |                         |                         |
|-----------------------|-------------------------|-------------------------|
| Mumbai - Maharashtra  | Nagpur - Maharashtra    | Pune - Maharashtra      |
| Panaji - Goa          | Bhopal - Madhya Pradesh | Indore - Madhya Pradesh |
| Raipur - Chhattisgarh |                         |                         |

### NW (9)

|                          |                         |                         |
|--------------------------|-------------------------|-------------------------|
| Delhi - Delhi            | Ahmedabad - Gujarat     | Surat - Gujarat         |
| Jaipur - Rajasthan       | Jodhpur - Rajasthan     | Lucknow - Uttar Pradesh |
| Varanasi - Uttar Pradesh | Chandigarh - Chandigarh | Amritsar - Punjab       |

### HIMALAYAN (6)

|                            |                        |                           |
|----------------------------|------------------------|---------------------------|
| Dehradun - Uttarakhand     | Nainital - Uttarakhand | Shimla - Himachal Pradesh |
| Srinagar - Jammu & Kashmir | Gangtok - Sikkim       | Jammu - Jammu & Kashmir   |

## Appendix B. Feature Matrix

| Feature               | Access                   | Prototype state |
|-----------------------|--------------------------|-----------------|
| Overview              | Public                   | Yes             |
| Forecast Explorer     | Public                   | Yes             |
| Weight Maps           | Public                   | Yes             |
| Weight Override       | Forecaster / Coordinator | Yes             |
| Skill & Verification  | Public                   | Yes             |
| Extreme Weather       | Public                   | Yes             |
| Alert Acknowledgement | Forecaster / Coordinator | Yes             |
| Email Alerts          | OTP subscriber           | Yes             |
| AAGAM Assistant       | Public                   | Yes             |
| Raw Data Query        | Public                   | Yes             |
| CSV/JSON Export       | Public                   | Yes             |
| Pipeline Health       | Forecaster / Coordinator | Yes             |
| Coordinator promotion | Coordinator              | Yes             |
| Public Event Share    | Public                   | Yes             |

### Closing note

This report is intended to accompany the project presentation and capture the full prototype lifecycle, feature set, operating model and technical foundations that cannot fit into a short slide deck.